Set 3: Embryology of the Eye & Optic Cup Derivatives

1. The very first structural indication of eye development appears in the early fourth week as a pair of shallow grooves on either side of the developing foregut brain. These are the:
a. Lens placodes
b. Optic vesicles
c. Optic pits (grooves)
d. Otic pits
e. Retinal crests
2. As the neural tube closes, the optic grooves expand laterally toward the surface ectoderm to form hollow, ball-like lateral outpocketings called the:
a. Optic cups
b. Optic stalks
c. Optic vesicles
d. Lens vesicles
e. Retinal discs
3. When the growing optic vesicle contacts the overlying surface ectoderm, it induces that ectoderm to thicken into a specialized structural disc called the:
a. Otic placode
b. Olfactory placode
c. Corneal epithelium
d. Lens placode
e. Optic disc placode
4. The optic vesicle subsequently invaginates or sinks inward at its distal pole, converting the single-walled sphere into a double-walled structure called the:
a. Lens vesicle
b. Optic stalk
c. Optic cup
d. Primitive vitreous cavity
e. Primary palate basin
5. The invagination of the optic cup does not just occur at the front; it also extends as a longitudinal groove along the inferior surface of the cup and its stalk. This groove is the:
a. Nasolacrimal groove
b. Optic (choroid/embryonic) fissure
c. Hyaloid sinus
d. Pupillary cleft
e. Foramen cecum canal
6. The optic or choroid fissure is highly important clinically because it permits the passage of which embryonic vascular vessel into the interior of the eye?
a. Central retinal artery
b. Ophthalmic artery
c. Hyaloid artery and vein
d. Anterior ciliary artery
e. Vortex venous trunk

7. Under normal conditions, the margins of the choroid fissure fuse completely during the seventh week of development, enclosing the hyaloid vessels inside the core of the:
a. Sclera
b. Lens vesicle
c. Cornea
d. Optic stalk (future optic nerve)
e. Ciliary body matrix
8. The double-walled optic cup gives rise to several intraocular structures. The outer, thinner layer of the optic cup differentiates into which permanent layer?
a. Sensory neural retina
b. Retinal pigment epithelium (RPE)
c. Vascular choroid coat
d. Fibrous sclera shell
e. Ciliary muscle body
9. The inner, thicker layer of the optic cup undergoes complex cellular differentiation to give rise to the posterior nine-tenths of the eye's nervous tissue, which is the:
a. Outer pigment epithelium
b. Vascular choroid
c. Sensory neural retina (containing rods, cones, bipolar, and ganglion cells)
d. Vitreous body face
e. Inner table of the sclera
10. The primitive space between the inner and outer layers of the optic cup gradually disappears as the two layers fuse, but it remains a potential line of structural weakness prone to:
a. Congenital cataracts
b. Glaucomatous cupping
c. Retinal detachment
d. Papilledema formation
e. Corectopia anomalies
11. The anterior portion of the optic cup does not form nervous tissue. Instead, both layers remain thin and grow forward to cover the ciliary body and the back of the iris, forming the:
a. Corneal endothelium
b. Iris stroma matrix
c. Pars ceca retinae (ciliary and iridial parts of the retina)
d. Suspensory zonules ring
e. Scleral spur fibers
12. The sphincter pupillae and dilator pupillae muscles of the iris are highly unique in embryology. Unlike almost all other muscles in the body, they develop from the ectoderm of the:
a. First pharyngeal arch mesenchyme
b. Neural crest cells exclusively
c. Optic cup (neuroectoderm)
d. Surface ectoderm placode
e. Paraxial mesoderm blocks

13. The lens placode invaginates into the surface ectoderm and eventually pinches off completely to form a hollow, fluid-filled sphere situated within the mouth of the optic cup. This is the:
a. Optic vesicle
b. Optic stalk
c. Optic groove
d. Lens vesicle
e. Secondary palate shell
14. The posterior cells of the lens vesicle elongate rapidly to form transparent, primary lens fibers that gradually grow forward to completely obliterate the internal cavity of the:
a. Optic cup
b. Lens vesicle
c. Vitreous chamber
d. Anterior chamber
e. Posterior chamber
15. Throughout life, secondary lens fibers continue to be added to the crystalline lens core. These secondary fibers develop continuously from the cells located in the:
a. Posterior pole of the lens
b. Central nucleus region
c. Equatorial zone (radial margins) of the lens
d. Anterior capsular epithelium only
e. Vitreous body face
16. While the lens fibers develop from ectoderm, the surrounding hyaline capsule of the lens is formed as a structural secretion of the:
a. Mesoderm cells of the orbit
b. Neural crest rows
c. Lens epithelium and its developing fibers
d. Ciliary processes epithelium
e. Choroid capillaries
17. The fibrous tunic (sclera) and the vascular tunic (choroid) of the eyeball develop from a dense condensation of cells that envelops the outside of the optic cup. These cells are a mix of:
a. Surface ectoderm and endoderm
b. Mesoderm and neural crest cells
c. Hypoblast and primitive streak rows
d. First arch and second arch branchial cells
e. Epiblast and yolk sac matrices
18. The cornea develops from multiple embryonic layers. Its outermost anterior epithelial lining is derived directly from the:
a. Optic cup neuroectoderm
b. Neural crest cells
c. Surface ectoderm
d. Paraxial mesoderm
e. Splanchnic mesoderm
19. The thick substantia propria (stroma) of the cornea and its inner endothelial lining develop from waves of migrating cells derived from the:
a. Surface ectoderm

b. Neural crest cells
c. Endoderm lining of the foregut
d. First branchial arch cartilage
e. Optic vesicle core

20. The anterior and posterior chambers of the eye develop as cleft-like spaces within the mesenchyme situated between the developing lens and the:

a. Sensory neural retina
b. Optic stalk axis
c. Primitive cornea
d. Vitreous body core
e. Scleral venous sinus border

21. In the fetus, the pupil is temporarily covered by a vascular connective tissue layer that normally disappears completely before birth. This transient embryonic layer is the:

a. Pupillary membrane
b. Tenon's capsule face
c. Suspensory zonule ring
d. Hyaloid arterial trunk
e. Scleral venous sinus meshwork

22. Persistence of segments of the pupillary membrane across the pupil path after birth can cause mild to moderate visual obstructions. This condition is termed pathologically as:

a. Aniridia
b. Persistent pupillary membrane (PPM)
c. Congenital aphakia
d. Microphthalmia
e. Cyclopia

23. The vitreous body develops in stages. The primary vitreous is vascularized and formed largely by the hyaloid vessels, whereas the permanent gelatinous secondary vitreous is avascular and produced by the:

a. Surface ectoderm cells
b. Inner layer of the optic cup and lens vesicle cells
c. Vascular choroid capillaries
d. Scleral osteoblasts
e. Extraocular recti muscle sheaths

24. The optic stalk connects the optic cup directly to the diencephalon. As the axons of the retinal ganglion cells grow backward through the wall of the stalk, it converts into the:

a. Oculomotor nerve
b. Trochlear nerve
c. Optic nerve (CN II)
d. Abducens nerve
e. Ophthalmic nerve

25. Myelination of the optic nerve fibers begins late in fetal development and is executed by oligodendrocytes. This process normally stops abruptly at which precise anatomical structure?

a. The optic chiasm
b. The optic tract exit

c. The lateral geniculate body
d. The optic disc (lamina cribrosa)
e. The equator of the globe

26. If the choroid/embryonic fissure fails to close completely during the seventh week, a notch-like defect occurs in the inferior segment of the ocular tissues. This congenital anomaly is:

a. Cataract
b. Aniridia
c. Coloboma (coloboma iridis/retinae)
d. Cyclopia
e. Microphthalmia

27. A coloboma can affect various structures along the inferior line of invagination. A typical coloboma of the iris is characterized by a keyhole-shaped defect located in which quadrant of the iris?

a. Superior quadrant
b. Lateral quadrant
c. Inferior quadrant
d. Medial quadrant exclusively
e. Central apex only

28. If the lens or its capsule becomes opaque during intrauterine development due to maternal infections like Rubella (German measles) during the 4th to 8th week, the child is born with:

a. Congenital glaucoma
b. Congenital cataract
c. Retinal detachment
d. Anophthalmia
e. Optic nerve atrophy

29. Congenital glaucoma (buphthalmos) is a severe birth defect characterized by a massive increase in intraocular pressure that stretches the infant's globe. It occurs due to an abnormal development or blockage of the:

a. Hyaloid canal path
b. Ciliary processes epithelium
c. Trabecular meshwork or scleral venous sinus (Canal of Schlemm)
d. Primary vitreous body cavity
e. Long ciliary arterial trunks

30. A rare, catastrophic developmental anomaly occurs when a failure of midline frontonasal development leads to complete fusion of the optic vesicles, resulting in a single midline orbit and eye. This is:

a. Aniridia
b. Microphthalmia
c. Cyclopia (synophthalmia)
d. Cryptophthalmos
e. Aphakia

31. Complete absence of the eyeball at birth due to a total failure of the optic vesicle to form or sprout from the diencephalon during the fourth week is:

a. Microphthalmia
b. Anophthalmia
c. Coloboma

d. Corectopia
e. Polycoria

32. A condition where the eyeball is abnormally small but structurally present, often due to intrauterine cytomegalovirus or toxoplasmosis infections arresting growth, is:
a. Macrophthalmia
b. Anophthalmia
c. Exophthalmos
d. Microphthalmia
e. Cryptophthalmos

33. Congenital aphakia refers to a complete structural absence of the lens inside the eye at birth. This anomaly stems from a total failure of induction or formation of the:
a. Optic cup inner layer
b. Lens placode (or failure of the lens vesicle to pinch off)
c. Primary vitreous matrix
d. Neural crest cell migration wave
e. Corneoscleral limbus rim

34. A complete congenital absence of the iris, often associated with mutations in the PAX6 gene, is a condition known as:
a. Aniridia
b. Coloboma
c. Polycoria
d. Corectopia
e. Synophthalmia

35. Cryptophthalmos is a rare condition where the eyelids fail to separate during development, leaving the skin of the face continuing uninterrupted completely over a hidden:
a. Nasal cavity
b. Oral vestibule path
c. Eyeball structure
d. External ear canal
e. Frontal sinus chamber

36. The hyaloid artery normally undergoes total degeneration in its distal segment during the late fetal period. If a remnant of this vessel persists as a non-functional strand extending from the optic disc through the vitreous, it is termed a:
a. Coloboma strand
b. Bergmeister's papilla (or persistent hyaloid artery remnant)
c. Persistent pupillary membrane loop
d. Tunica vasculosa lentis patch
e. Mittendorf dot exclusively

37. A small, opaque white spot located on the posterior capsule of the lens representing the exact attachment point of the embryonic hyaloid artery is clinically named a:
a. Mittendorf dot
b. Koplik spot
c. Hutchinson notch
d. Fordyce spot
e. Epibulbar dermoid

38. The optic vesicle induces the surface ectoderm to form the lens. If the optic vesicle is surgically removed or fails to reach the ectoderm during the fourth week, what occurs?
a. A normal lens still forms independently
b. No lens forms, confirming that the optic vesicle is a mandatory inductor tissue
c. The ectoderm transforms into a second optic nerve
d. The cornea develops inside the brain
e. The ciliary body hyperplasiates massively
39. The neural retina is structurally continuous with the brain itself. Anatomically, the optic nerve is not a true peripheral nerve, but rather a modified tract of the central nervous system covered by:
a. Epineurium and perineurium sheaths
b. Simple skeletal muscle fascia
c. The three meningeal layers (dura, arachnoid, and pia mater)
d. Respiratory mucosal cells
e. Subcutaneous fat sheets
40. Because the subarachnoid space extends around the optic nerve up to the back of the globe, any rise in intracranial pressure is directly transmitted around the nerve to produce:
a. Glaucomatous cupping of the disc
b. Edema of the optic disc (papilledema)
c. Complete detachment of the iris
d. Rupture of the cornea stroma
e. Prolapse of the lens vesicle fibers
41. The eyelids develop from small folds of surface ectoderm containing a core of mesenchyme that grow toward each other during the seventh week. These folds fuse completely and remain stuck together until approximately which month of pregnancy?
a. Third month
b. Fifth month
c. Seventh or eighth month
d. They never fuse during normal development
e. First week postnatal
42. The skeletal muscles that move the eyeball (extraocular muscles) develop from the mesenchyme of the preotic somites, which are innervated by cranial nerves:
a. V, VII, and IX
b. III, IV, and VI
c. I, II, and VIII
d. IX, X, and XI
e. VII, IX, and XII
43. The lacrimal gland develops as a series of solid buds that sprout from the surface ectoderm located at which relative position of the conjunctival sac?
a. Upper medial corner
b. Lower medial corner
c. Upper lateral corner
d. Lower lateral corner
e. Absolute posterior center apex

44. The retinal pigment epithelium layer becomes pigmented early in development, making the eye one of the first colored structures visible in the embryo. This pigmentation begins during the:
a. Second week
b. Fifth to sixth week
c. Fifth month
d. Ninth month
e. Day of birth
45. Cells of the neural crest contribute heavily to the eye. Which of the following intraocular structures is entirely derived from neural crest cells?
a. Sensory neural retina
b. Crystalline lens fibers
c. Stroma of the iris, ciliary body, and choroid
d. Retinal pigment epithelium
e. Anterior epithelial lining of the cornea
46. The double-walled optic cup communicates with the brain cavity via the lumen of the optic stalk. This lumen is gradually obliterated as which fiber type fills the channel?
a. Connective tissue scars
b. Axons of retinal ganglion cells growing toward the brain
c. Hyaloid arterial smooth muscle cells
d. Oligodendrocyte myelin plates
e. Ependymal cell rows
47. If the lens vesicle fails to detach completely from the surface ectoderm during the fifth week, it results in a congenital central clouding of the cornea and lens known as:
a. Peters anomaly (anterior segment dysgenesis)
b. Buphthalmos
c. Microphthalmia
d. Aniridia
e. Coloboma of the macula
48. The tunica vasculosa lentis is an embryonic vascular network that wraps around the developing lens. It normally undergoes complete regression; its persistence and subsequent fibrosis in premature infants exposed to high oxygen therapy is called:
a. Retinoblastoma
b. Retinopathy of prematurity (retrolental fibroplasia)
c. Congenital aphakia
d. Buphthalmos
e. Optic atrophy
49. The outer layer of the optic cup remains a single layer of cuboidal cells, while the inner layer stratifies into multiple layers. This structural divergence represents the process of:
a. Gastrulation
b. Neurulation
c. Cellular differentiation of the retinal sheets
d. Blastocyst hatching
e. Pharyngeal arch segmentation

50. The definitive scleral venous sinus (Canal of Schlemm) develops as a specialized endothelial-lined channel that tracks the circumference of the iridocorneal angle, arising from local:

a. Optic cup neuroectodermal loops
b. Neural crest cell matrices
c. Vascular mesodermal plexuses at the corneoscleral junction
d. Lens vesicle equatorial outgrowths
e. Surface ectoderm invaginations

Answer Key Table

Q	**Q**	**Q**	**Q**	**Q**	**Q**	**Q**

1-8	9-16	17-24	25-32	33-40	41-48	49-50
e	c	b	d	c	b	c
c	c	c	c	a	b	c
d	c	b	b	a	c	
c	b	c	c	c	b	
b	b	c	c	c	c	
c	c	a	d	b	b	
d	c	c	b	a	a	
b	b	c	d	b	b	